

Documentation for Task 2

See first - All files are on climb:

[shared-team/people/antony/rp1b](#)

True randomisation of mutation reads can be enabled by editing removing `set.seed(123)` in `functions/make_mutations.py`

The full data for simulated ecoli fasta -> fastq reads, including output vcf, bai, bam files

Part 1: Manually simulating reads, mapping and variant calling

Note - Can substitute with the `simulate_mutations.py` & `simulate_reads.py` Pipeline

Simulating reads from reference genome

Running cell 1 in `part1_manual_pipeline.ipynb`

On test file located in `/data/raw`:

`EcoliK12-MG1655.fasta`

Note: many input arguments are hardcoded, for example for 300 SNPs and 10 INDELs, but corresponding variables are modifiable in this Jupyter Notebook.

Running minimap2 and bcftools on .fasta and .fastq files

`{fasta_name}` = eg. `EcoliK12-MG1655`

`{fastq_name}` = eg. `NC_000913.3`

`{bam_output_name}` = eg. `mapped_reads`

`{vcf_output_name}` = eg. `variants`

In Bash (inside directory of `part1 manual pipeline.ipynb`):

Make sure to first `conda create`, `conda install` **`minimap2`**, **`samtools`**, and **`bcftools`**, and `conda activate`

`minimap2 -a -x sr /data/raw/{fasta_name}.fasta /data/processed/{fastq_name}.fastq | samtools view -h -F 0x900 - | samtools sort -O bam > {bam_output_name}.bam`

`bcftools mpileup -Ou -f {fasta_name}.fasta {bam_output_name}.bam | bcftools call -vc -Ov > {vcf_output_name}.vcf`

Explanation of Pipeline

Whilst not a continuous “|” piped pipeline, the following sequential steps in the pipeline should yield repeatable outputs.

Bash scripts to run simulated mutations and reads pipeline:

Mutation simulation → vcf is achieved by 5 steps running each .py file, in order of:

1. simulate_mutations.py
2. simulate_reads.py
3. run_bcftools_only.py
4. run_snippy_only.py
5. merge_vcfs.py

Using argparse to allow tailored simulated reads, and alignment and variant calling.

Instructions to follow below for details:

1. **Initialise intermediary mutated file (this allows use of .txt file separating randomised simulate_mutations.py and non-randomised simulate_reads.py so simulate_reads.py)**

INTERMEDIARY_FILE="mutated_sequence.txt"

Remember to choose different INTERMEDIARY_FILE name if duplicate name already present in file. Name must be unique.

2. **Run the first script to create the intermediary file**

```
python simulate_mutations.py EcoliK12-MG1655.fasta \  
--output-seq "$INTERMEDIARY_FILE" \  
--output-gt "ground_truth.txt" \  
--snps 300 --indels 10
```

(--snps 300 --indels 10) is optional as these are already defaults, but allows modification

3. **Run the second script to read the intermediary file and complete the process**

```
python simulate_reads.py NC_000913.3 \  
--input-seq "$INTERMEDIARY_FILE" \  
--depth 30 --read_len 100
```

This generates a FASTQ file, e.g., 'data/processed/NC_000913.fastq'

(--depth 30 --read_len 100) is optional as these are already defaults, but allows modification

Run with bcftools alone

*Again, make sure to first conda create, conda install **minimap2**, **samtools**, and **bcftools**, and conda activate.*

```
python run_bcftools_only.py EcoliK12-MG1655.fasta NC_000913.3.fastq
```

This reads the FASTA and FASTQ, and produces the BAM and VCF files.

You can now calculate its precision/recall – see section What is the precision/recall of the SNP caller?

Run with snippy alone

*Conda create package **snippy**:*

```
conda create --name snippy_460 snippy=4.6.0
```

```
conda activate snippy_460
```

```
python run_snippy_only.py \  
--reference EcoliK12-MG1655.fasta \  
--read1 NC_000913.3.fastq \  
--output_prefix ecoli_snippy
```

(--read2 sample_A_R2.fastq \) is optional, to accommodate single-ended data with option for paired-end data

You can now calculate its precision/recall – see section What is the precision/recall of the SNP caller?

Merge bcftools and snippy with output combined into singular vcf using bcftools merge

Files are compressed with bgzip, indexed with tabix and bcftools merge is applied:

Example file 1 = ecoli_snippy.raw.vcf

Example file 2 = EcoliK12-MG1655.vcf

Output file = merged_samples.vcf.gz

Example Execution

```
python merge_vcfs.py data/processed/EcoliK12-MG1655.vcf data/snippy_only/ecoli_snippy.raw.vcf -  
o data/processed/merged_samples.vcf.gz
```

Unzip with gunzip, eg.:

```
gunzip -k data/processed/merged_samples.vcf.gz
```

Ensure to conda activate to environment with bcftools package

Make sure to specify correct directory locations of vcf files. More than two vcf files can be merged.

You can now calculate its precision/recall – see section What is the precision/recall of the SNP caller?

Check that output is located in...

Directory:

/data/processed

What is the precision/recall of the SNP caller?

In the **functions/making_mutations.py**, argument `output_gt_path` the filename **"ground_truth.txt"** is required. Function **write_ground_truth** writes the `ground_truth.txt` file which is a tab delimited file containing the "CHROM", "POS", "REF", "ALT", "TYPE", "LEN" of each mutation.

calculate_metrics.py uses the **"ground_truth.txt"** file to calculate precision and recall.

Here, precision and recall are defined based on: TP=True positive, FP=False positive, FN=False negative, Recall=Sensitivity, Precision=Positive predictive value

Precision (PPV): $TP / (TP + FP)$

Recall (Sensitivity): $TP / (TP + FN)$

In Bash enter code:

```
python calculate_metrics.py check_intermediary_file/"ground_truth.txt"  
data/processed/{variant_called_filename}.vcf
```

Replace {variant_called_filename} with relevant vcf file from bcftools_only

Discussion of Pipeline

Comments on my SNP and indel producing code:

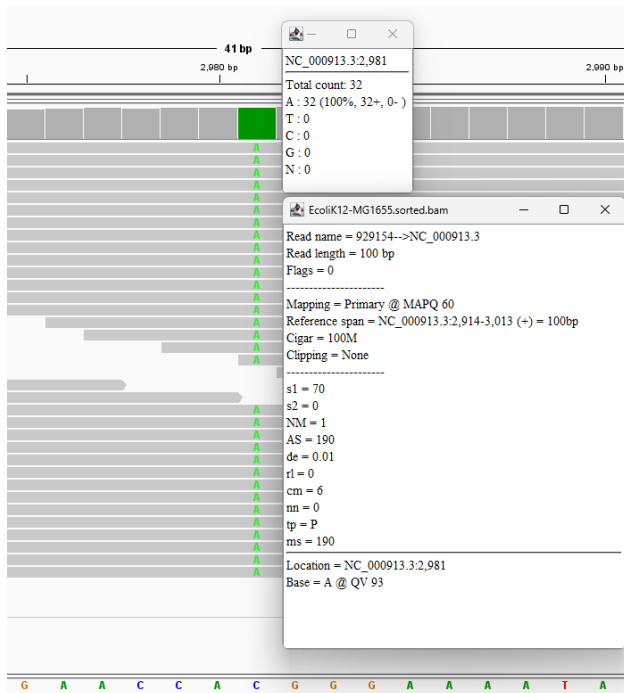
Overall, my code was successful in producing SNPs and indels using the 4 functions."{function}".py files.

Running these function files through `simulate_mutations.py` and `simulate_reads.py` outputs the vcf, bam, bai files required for viewing in IGV for sanity checks.

In IGV:

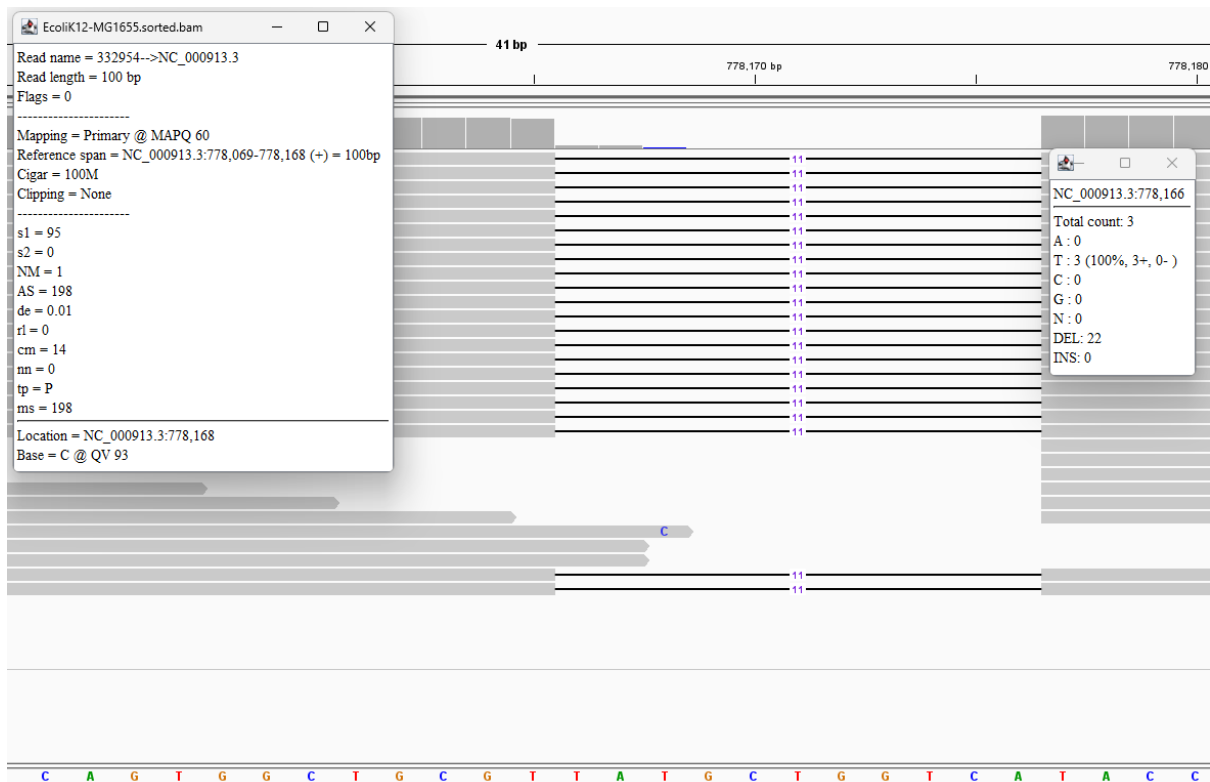
Check SNP from vcf

NC_000913.3 2981 . C A



Check indel from vcf (deletion)

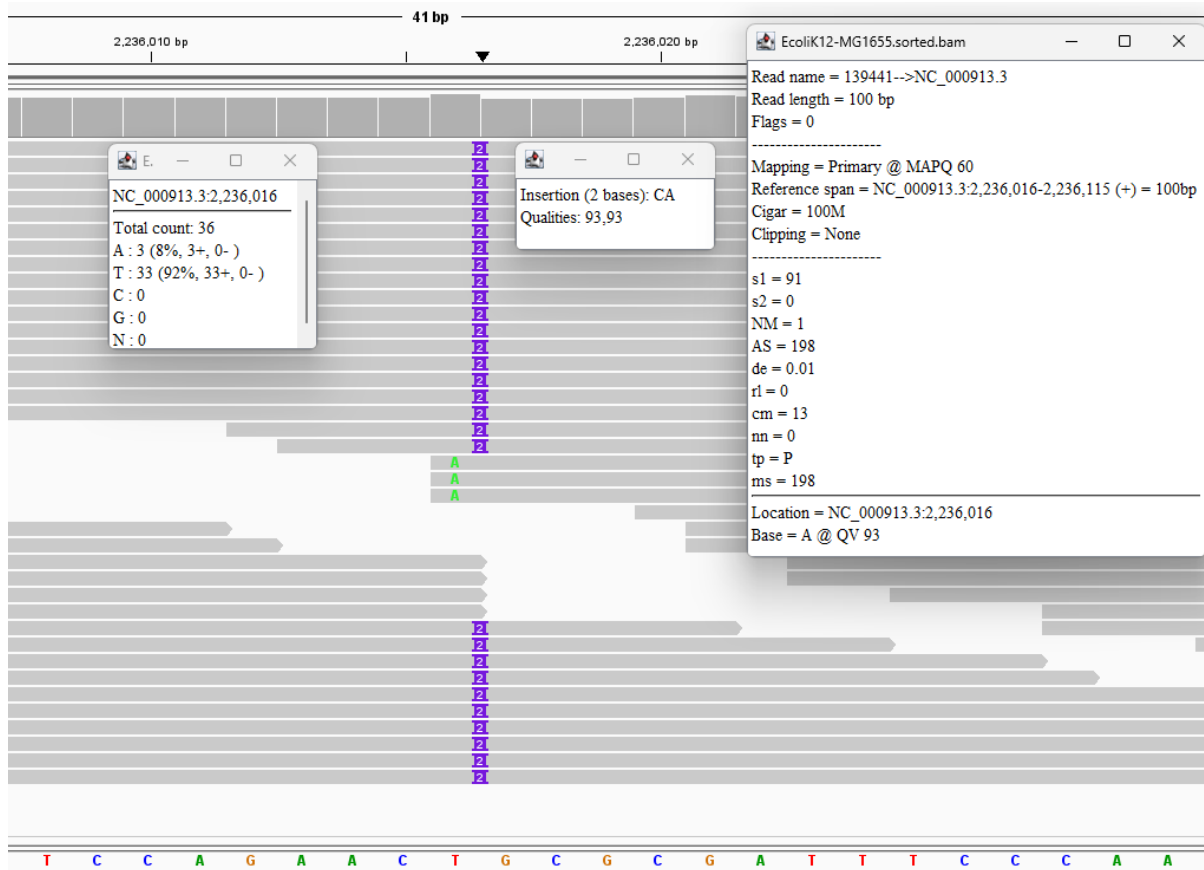
NC_000913.3 778165 . TTATGCTGGTCATA TTA 166.459 . INDEL



Looks like a deletion as coverage is significantly lower uniformly over 11 bases, and read overhang contains 'C' mismatch which could indicate a mis-mapped read.

Check indel from vcf (insertion)

NC_000913.3 2236016 . T TCA 170.458 . INDEL



Mostly uniform insertion of 2 bases. 3 possible mis-mapped reads (all uniform)

Comments on Precision and Recall:

For bcftools of the data/processed/EcoliK12-MG1655.vcf against ground truth:

```
(my_env) jovyan:/shared/team/people/antony/rp1b$ python calculate_metrics.py check_intermediary_file/ground_truth.txt data/processed/EcoliK12-MG1655.vcf
Parsing ground truth file: check_intermediary_file/ground_truth.txt...
Loaded 310 ground truth variants.
Comparing VCF file: data/processed/EcoliK12-MG1655.vcf against ground truth...
```

```
=====
Variant Caller Performance Metrics
=====
```

```
Confusion Matrix Components:
True Positives (TP): 299
False Positives (FP): 7
False Negatives (FN): 11
```

```
Total Counts:
Total GT Variants: 310 (TP + FN)
Total Called Variants: 306 (TP + FP)
```

```
Final Scores:
Recall (Sensitivity): 0.9645
Precision (PPV): 0.9771
F1 Score: 0.9708
=====
```

Recall (Sensitivity) = 0.9645

Precision (PPV) = 0.9771

For snippy of the data/snippy_only/ecoli_snippy.raw.vcf against ground truth:

```
python calculate_metrics.py check_intermediary_file/ground_truth.txt
data/snippy_only/ecoli_snippy.raw.vcf
```

```
(snippy_460) jovyan:/shared/team/people/antony/rp1b$ python calculate_metrics.py check_intermediary_file/ground_truth.txt data/snippy_only/ecoli_snippy.raw.vcf
Parsing ground truth file: check_intermediary_file/ground_truth.txt...
Loaded 310 ground truth variants.
Comparing VCF file: data/snippy_only/ecoli_snippy.raw.vcf against ground truth...
```

```
=====
Variant Caller Performance Metrics
=====

Confusion Matrix Components:
True Positives (TP): 296
False Positives (FP): 10
False Negatives (FN): 14

Total Counts:
Total GT Variants: 310 (TP + FN)
Total Called Variants: 306 (TP + FP)

Final Scores:
Recall (Sensitivity): 0.9548
Precision (PPV): 0.9673
F1 Score: 0.9610
=====
```

Recall (Sensitivity) = 0.9548

Precision (PPV) = 0.9673

Combined vcf

```
python calculate_metrics.py check_intermediary_file/ground_truth.txt
data/processed/merged_samples.vcf
```

```
(my_env) jovyan:/shared/team/people/antony/rp1b$ python calculate_metrics.py check_intermediary_file/ground_truth.txt data/processed/merged_samples.vcf
Parsing ground truth file: check_intermediary_file/ground_truth.txt...
Loaded 310 ground truth variants.
Comparing VCF file: data/processed/merged_samples.vcf against ground truth...
```

```
=====
Variant Caller Performance Metrics
=====

Confusion Matrix Components:
True Positives (TP): 299
False Positives (FP): 7
False Negatives (FN): 11

Total Counts:
Total GT Variants: 310 (TP + FN)
Total Called Variants: 306 (TP + FP)

Final Scores:
Recall (Sensitivity): 0.9645
Precision (PPV): 0.9771
F1 Score: 0.9708
=====
```

Recall (Sensitivity) = 0.9645

Precision (PPV) = 0.9771

Comparison of bcftools, snippy, combined vcf: Precision and Recall

The combined vcf had highest recall and precision, as it culminated unique rows of mutation data from both variant callers.

Recall: bcftools had higher recall (0.9645) than snippy (0.9548), with a difference of 0.0097

Precision: bcftools had higher precision (0.9771) than snippy (0.9673), with a difference of 0.0098

Comment on real e coli data output Precision and Recall

Testing on real e coli data located in path:

data/processed/SRR25083113_1.fastq

https://www.ncbi.nlm.nih.gov/datasets/genome/GCF_000005845.2/

Taxon: Escherichia coli str. K-12 substr. MG16552/

Confirmation of extremely high/low precision & recall reads with tview

```
samtools tview EcoliK12-MG1655.sorted.bam ../raw/ EcoliK12-MG1655.fasta
```

[illegible]

Deletion at 778165 can also be noted on tview. The three overhanging reads seen here confirm the three overhanging reads seen on IGV.

Documentation of all versions of packages used

bcftools 1.21

Using htplib 1.21

samtools 1.20

snippy 4.6.0